

Problem

Rework Prob. 3.5 using mesh analysis.

Step-by-step solution

Step 1 of 4

Refer to the circuit diagram in Figure 3.54 in the textbook.

Redraw the circuit diagram with node identification.

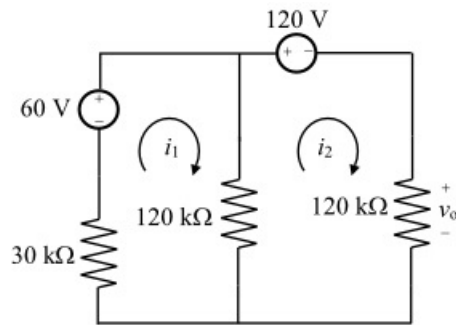


Figure 1

Apply Kirchhoff's voltage law around loop 1.

$$30 \times 10^3 i_1 - 60 + 120 \times 10^3 (i_1 - i_2) = 0$$

$$150 \times 10^3 i_1 - 120 \times 10^3 i_2 = 60 \quad \text{..... (1)}$$

Apply Kirchhoff's voltage law around loop 2.

$$120 \times 10^3 i_2 + 120 \times 10^3 (i_2 - i_1) + 120 = 0$$

$$-120 \times 10^3 i_1 + 240 \times 10^3 i_2 = -120 \quad \text{..... (2)}$$

Step 2 of 4

Solve equation (1) and (2).

Multiply equation (1) by 2.

$$300 \times 10^3 i_1 - 240 \times 10^3 i_2 = 120 \quad \text{..... (3)}$$

Step 3 of 4

Add equations (3) and (2).

$$\begin{aligned} 300 \times 10^3 i_1 - 240 \times 10^3 i_2 &= 120 \\ \underline{-120 \times 10^3 i_1 + 240 \times 10^3 i_2} &= \underline{-120} \\ 180 \times 10^3 i_1 &= 0 \\ i_1 &= 0 \text{ A} \end{aligned}$$

Solve for current i_2 .

$$\begin{aligned} i_2 &= \frac{-120}{240 \times 10^3} \\ &= -0.5 \text{ mA} \end{aligned}$$

Step 4 of 4

Voltage across $120 \text{ k}\Omega$ resistor is,

$$\begin{aligned} v_o &= 120 \times 10^3 i_2 \\ &= 120 \times 10^3 (-0.5 \times 10^{-3}) \\ &= -60 \text{ V} \end{aligned}$$

Thus, voltage v_o across $120 \text{ k}\Omega$ resistor is -60 V .